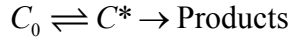


Supporting Information

I. A Simple Lindemann Model

A simple Lindemann model gives a very different result from the microcanonical argument given in the Introduction. Classifying the energy levels of C above the reaction barrier E_0 as a distinct species C^* , and those below as C_0 , a unimolecular reaction can be written in two steps:



In the high-pressure limit, the collisional activation rate of C_0 and the collisional deactivation rate of C^* are both fast and nearly equal; therefore, all the energy levels of C have Boltzmann populations, so $[C^*]/[C] = Q_{C^*}/Q_C$ where the partition functions are given by

$$Q_{C^*} = \int_{E_0}^{\infty} \rho_C(E) e^{-E/k_B T} dE = e^{-E_0/k_B T} \int_0^{\infty} \rho_C(E + E_0) e^{-E/k_B T} dE$$

and $Q_C = \int_0^{\infty} \rho_C(E) e^{-E/k_B T} dE$

To maintain this population of C^* , the collisional activation rate must be greater than or equal to the overall reaction rate for $C \rightarrow \text{Products}$:

$$\text{Collisional Activation Rate} > k_{\infty}(T)[C]$$

which will be satisfied if the nearly equal but slightly smaller collisional deactivation rate is greater than the reaction rate:

$$\beta k_s [M][C^*] > k_{\infty}(T)[C]$$

that is, if:

$$\begin{aligned} \beta k_s [M] &> k_{\infty}(T) \frac{[C]}{[C^*]} \\ \beta k_s [M] &> \left(\frac{k_B T}{h} \frac{Q_{TS}}{Q_C} e^{-E_0/k_B T} \right) \left(\frac{Q_C}{Q_{C^*}} \right) \\ \beta k_s [M] &> \left(\frac{k_B T}{h} \right) \int_0^{\infty} \rho_{TS}(E) e^{-E/k_B T} dE \bigg/ \int_0^{\infty} \rho_C(E + E_0) e^{-E/k_B T} dE \end{aligned}$$

At very low T , the right-hand side is very small, and the reaction would be expected to be in the high-pressure limit even at very low pressures, consistent with the simple microcanonical argument

given in the Introduction. However at high temperatures, the right-hand side depends on both temperature and molecular size in a rather complicated way. This is due to the fact that both integrands peak around $E = mk_B T$, where m is the number of active degrees of freedom, and E can be very large, even larger than E_0 . In this case, the integrals could be comparable, and then this model would predict very high pressures would be required to achieve the high-pressure limit.

II. The Derivations of Equation 10 and 12

Starting from Eq.(6):

$$\int_{E_{AB}}^{\infty} \rho_C(E) k_C(E) \phi(E) e^{-E/k_B T} dE = \frac{1}{1+\varepsilon} \int_{E_{AB}}^{\infty} \rho_C(E) k_C(E) e^{-E/k_B T} dE$$

and substituting the approximations from Eqs.(7)-(8) in Eq.(6) gives

$$\begin{aligned} & \int_{E_{AB}}^{\infty} \rho_C(E) \left\{ A_i \frac{\rho_C(E - E_{AB})}{\rho_C(E)} H_{E_{AB}} \right\} \left\{ \frac{\beta k_s[M]}{\beta k_s[M] + \sum_i A_i \frac{\rho_C(E - E_i)}{\rho_C(E)} H_{E_i}} \right\} e^{-E/k_B T} dE \\ &= \frac{1}{1+\varepsilon} \int_{E_{AB}}^{\infty} \rho_C(E) \left\{ A_i \frac{\rho_C(E - E_{AB})}{\rho_C(E)} H_{E_{AB}} \right\} e^{-E/k_B T} dE \end{aligned}$$

Simplifying the expression above yields

$$\int_{E_{AB}}^{\infty} \rho_C(E - E_i) \left\{ \frac{\beta k_s[M]}{\beta k_s[M] + \sum_i A_i \frac{\rho_C(E - E_i)}{\rho_C(E)} H_{E_i}} \right\} e^{-E/k_B T} dE = \frac{1}{1+\varepsilon} \int_{E_{AB}}^{\infty} \rho_C(E - E_i) e^{-E/k_B T} dE$$

Using the density of vibrational-rotational states given by Eq.(9)

$$\begin{aligned}
& \int_{E_{AB}}^{\infty} \left\{ \frac{(E - E_i + Z)^{m-1}}{\Gamma(m) \prod_i^s h\nu_i} \prod_i^r \left(\frac{8\pi^3 I_i}{h^2} \right)^{1/2} \right\} \left\{ \frac{\beta k_s [M]}{\beta k_s [M] + \sum_i A_i \left(\frac{E - E_i + Z}{E + Z} \right)^{m-1} H_{E_i}} \right\} e^{-E/k_B T} dE \\
&= \frac{1}{1 + \varepsilon} \int_{E_{AB}}^{\infty} \left\{ \frac{(E - E_i + Z)^{m-1}}{\Gamma(m) \prod_i^s h\nu_i} \prod_i^r \left(\frac{8\pi^3 I_i}{h^2} \right)^{1/2} \right\} e^{-E/k_B T} dE
\end{aligned}$$

Simplifying the expression above gives

$$\int_{E_{AB}}^{\infty} (E - E_i + Z)^{m-1} \left\{ \frac{\beta k_s [M]}{\beta k_s [M] + \sum_i A_i \left(\frac{E - E_i + Z}{E + Z} \right)^{m-1} H_{E_i}} \right\} e^{-E/k_B T} dE = \frac{1}{1 + \varepsilon} \int_{E_{AB}}^{\infty} (E - E_i + Z)^{m-1} e^{-E/k_B T} dE$$

This equation may be condensed even further by making the change in variable $w = (E + Z - E_{AB})/((m - 1)k_B T)$, where $Z = hc\langle\omega\rangle s/2$. Accordingly $H_{E_i} = H((m - 1)wk_B T + E_{AB} - Z - E_i)$ and $dE = ((m - 1)k_B T) dw$; therefore, we obtain

$$\begin{aligned}
& \int_a^{\infty} w^{m-1} \left\{ \frac{\beta k_s [M]}{\beta k_s [M] + \sum_i A_i \left(\frac{w(m-1)k_B T + E_{AB} - E_i}{w(m-1)k_B T + E_{AB}} \right)^{m-1} H_{E_i}} \right\} e^{-\{w(m-1)k_B T + E_{AB} - Z\}/k_B T} ((m-1)k_B T) dw \\
&= \frac{1}{1 + \varepsilon} \int_a^{\infty} w^{m-1} e^{-\{w(m-1)k_B T + E_{AB} - Z\}/k_B T} ((m-1)k_B T) dw
\end{aligned}$$

where $a = Z/(m-1)k_B T = (hc\langle\omega\rangle/2k_B T)(s/(m - 1)) \sim hc\langle\omega\rangle/2k_B T$. Simplifying the above expression yields

$$\int_a^{\infty} \frac{w^{m-1} e^{-(m-1)w}}{1 + \frac{1}{\beta k_s [M]} \sum_i A_i H_{E_i} \left[1 - \frac{E_i}{w(m-1)k_B T + E_{AB}} \right]^{m-1}} dw = \frac{1}{1 + \varepsilon} \int_a^{\infty} w^{m-1} e^{-(m-1)w} dw$$

which is Eq.(10). Approximating both integrands by simple function evaluations at $w = (1 + a^4)^{1/4}$ as described in the main text gives

$$\frac{(1+a^4)^{(m-1)/4} e^{-(m-1)(1+a^4)^{1/4}}}{1 + \frac{1}{\beta k_s [M]} \sum_i A_i H_{E_i} \left[1 - \frac{E_i}{(1+a^4)^{1/4} (m-1) k_B T + E_{AB}} \right]^{m-1}} (\Delta w) = \frac{1}{1 + \varepsilon} (1+a^4)^{(m-1)/4} e^{-(m-1)(1+a^4)^{1/4}} (\Delta w)$$

where Δw is the width of the sharp peak. Simplifying the above expression finally yields

$$\sum_i A_i H_{E_i} \left(1 - \frac{E_i}{E_{AB} + (m-1)(1+a^4)^{1/4} k_B T} \right)^{m-1} \cong \varepsilon \beta k_s [M] = \frac{\varepsilon \beta k_s P}{RT}$$

which is Eq.(12) as promised.

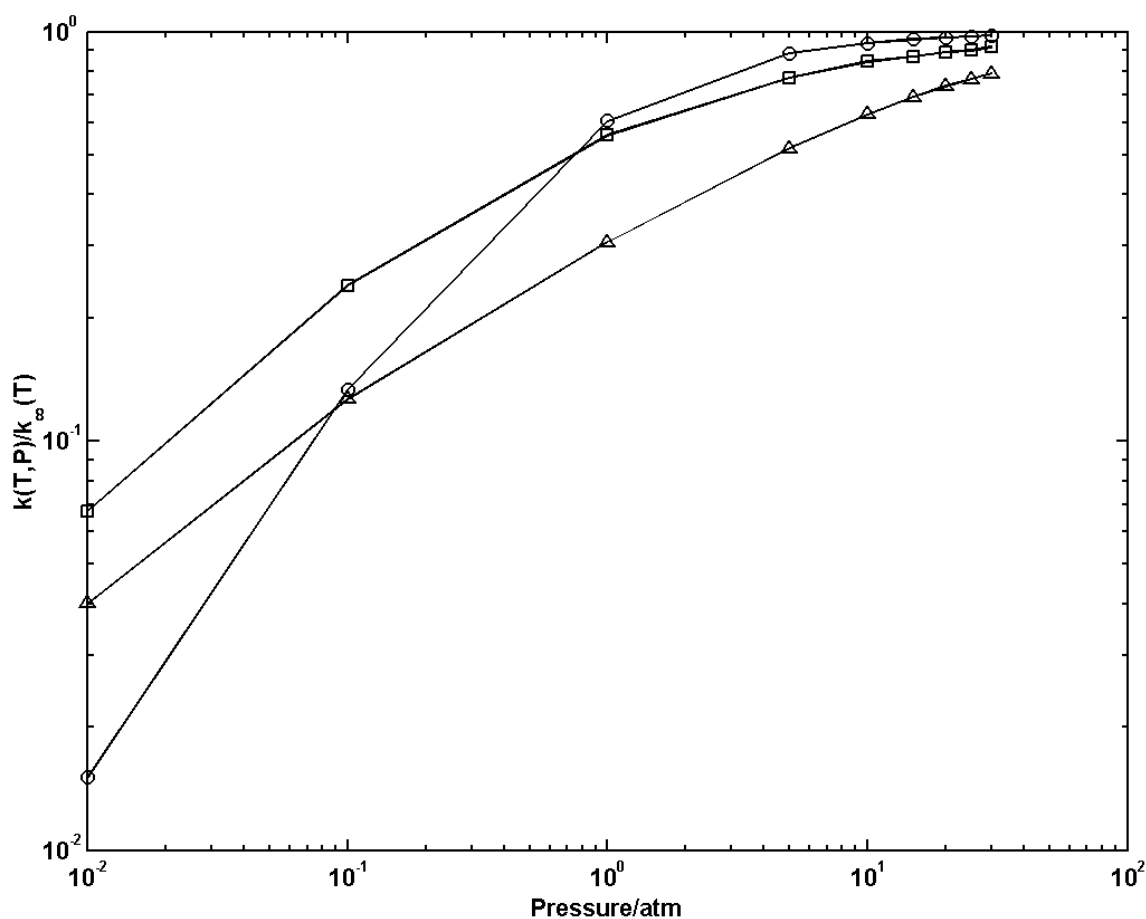


Figure 4. Comparison of the predicted ratio $k(T,P)/k_{\infty}(T)$ at $T = 1000$ K for the reaction $\text{C}_6\text{H}_5 + \text{C}_2\text{H}_2 \rightarrow \text{C}_6\text{H}_5\text{CHCH}$; $\square$ = ratios predicted by MultiWell; Δ = ratios predicted by CHEMDIS; $\circ$ = ratios predicted by Eq.(4) using Eq.(12) to compute $\alpha(T,P)$. This is a chemically activated system with several product channels. The modest discrepancy between MultiWell (i.e. master equations) and CHEMDIS (i.e. modified strong collision approximation) is comparable to that seen in other systems^{2,11-13}. The present approximation Eq.(12) is accurate when stabilization is a major channel.

The following plots give comparisons of switchover pressures [pressures where $k(T,P)/k_\infty(T) = 1/2$] predicted by the approximations in this present work (Eq.(12)) and the contours of $k(T,P)/k_\infty(T)$ using more detailed calculations by CHEMDIS. In the region of (T,P) space above the P_{switch} curve, the rate constant can be approximated by the high-pressure limit. The contour lines show our approximate calculation for P_{switch} accurately locates the “transition regime” (gray area) for $k(T,P)$ where the rate constant turns from $k_\infty(T)$ into a pressure-dependent rate constant as temperature increases. The required A ’s and E_a ’s for all the channels listed for each reaction were taken from literature sources indicated in Table 1. Other CHEMDIS parameters are the same as described in the text. The densities of states required by CHEMDIS come from the heat capacities fitted to the three-frequency form proposed by Bozzelli et al. using their THERFIT software²⁰.

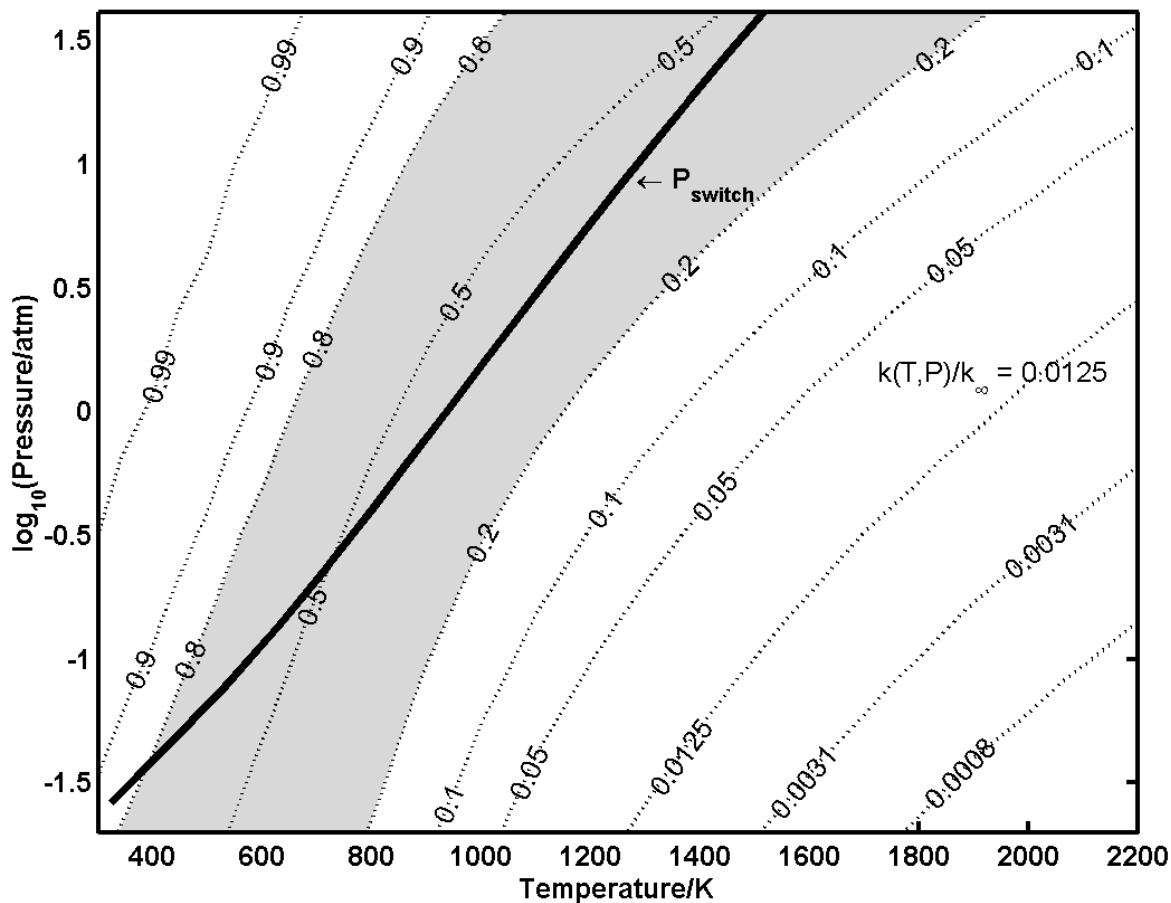


Figure 5. $\text{C}_6\text{H}_5 + \text{C}_2\text{H}_2 \rightarrow \text{C}_6\text{H}_5\text{CHCH}$ (multiple-channel chemically-activated system); switchover pressures for $k(T,P)/k_\infty(T) = 1/2$ predicted by Eq.(12) vs. CHEMDIS $k(T,P)/k_\infty(T)$ contours.

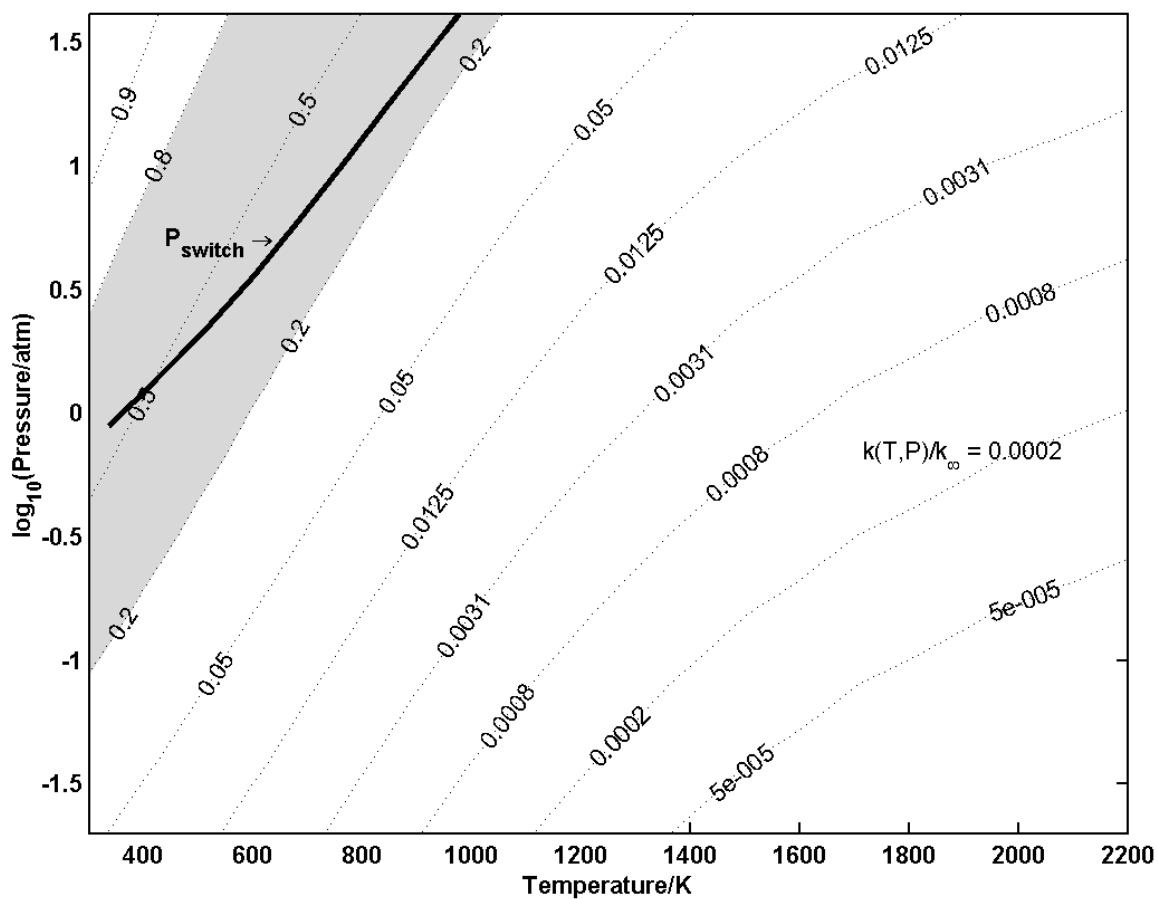


Figure 6. $\text{CH}_3\text{CH}_2\text{O} \rightarrow \text{CH}_2\text{O} + \text{CH}_3$; switchover pressures for $k(T,P)/k_{\infty}(T) = 1/2$ predicted by Eq.(12) vs. CHEMDIS $k(T,P)/k_{\infty}(T)$ contours.

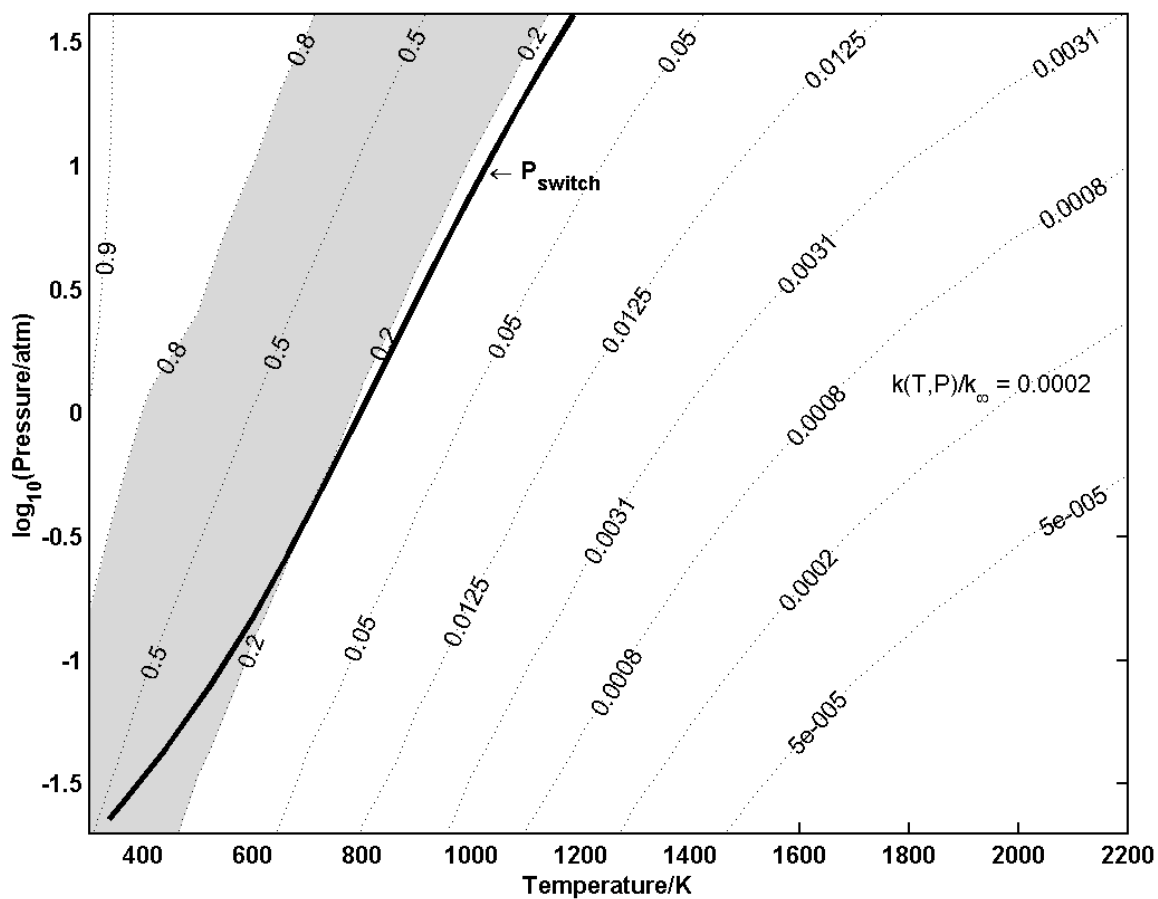


Figure 7. $\text{C}_2\text{H}_5\text{OO} \rightarrow \text{C}_2\text{H}_5 + \text{O}_2$; switchover pressures for $k(T,P)/k_{\infty}(T) = 1/2$ predicted by Eq.(12) vs. CHEMDIS $k(T,P)/k_{\infty}(T)$ contours.

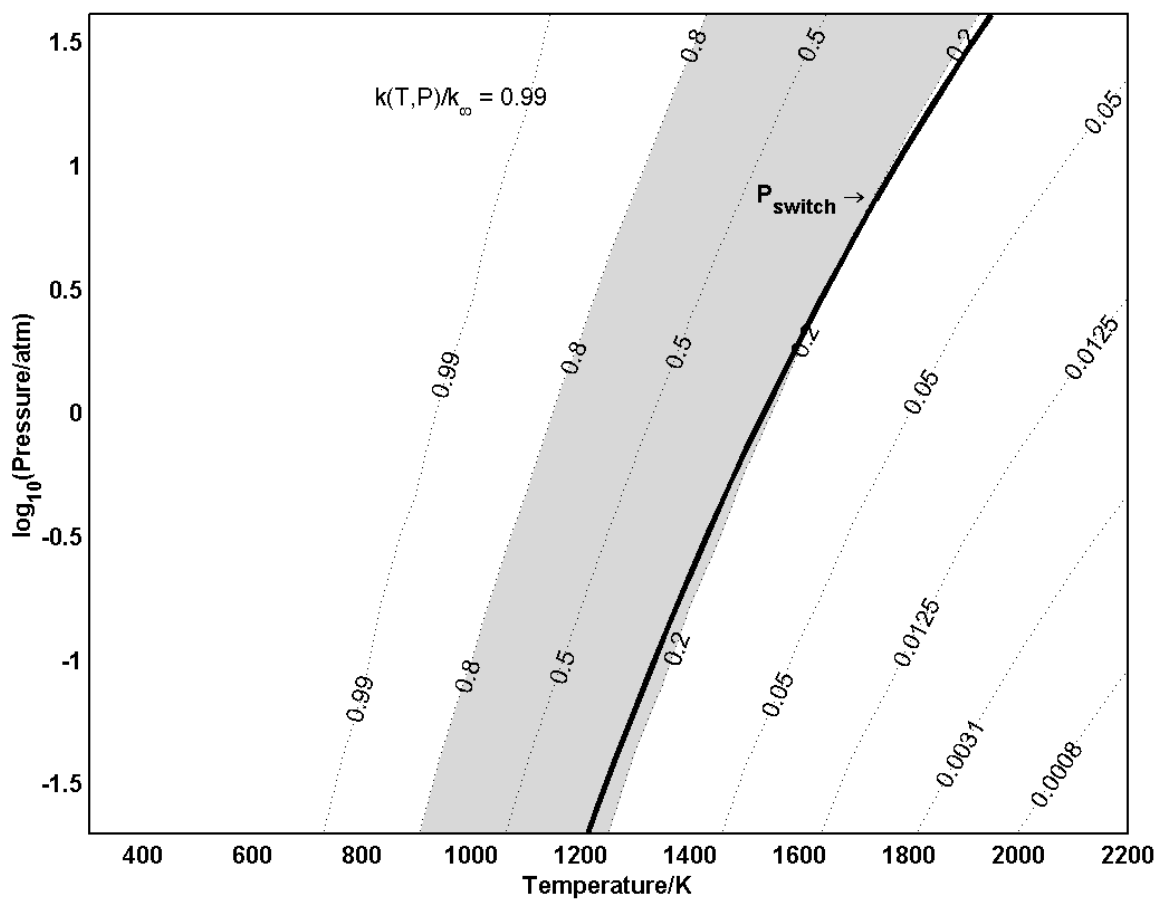


Figure 8. $\text{C}_4\text{H}_{10} \rightarrow 2 \text{C}_2\text{H}_5$; switchover pressures for $k(T,P)/k_{\infty}(T) = 1/2$ predicted by Eq.(12) vs. CHEMDIS $k(T,P)/k_{\infty}(T)$ contours.

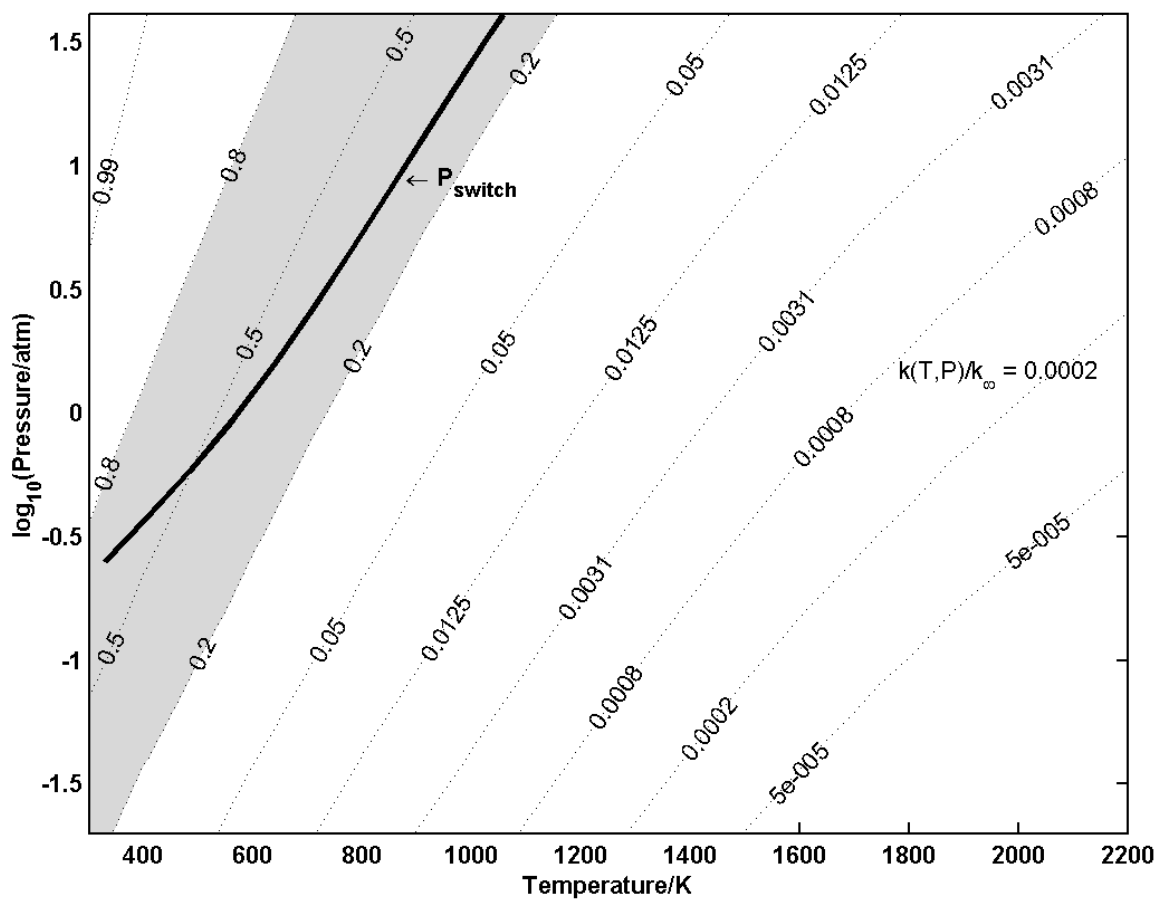


Figure 9. $\text{CH}_3\text{OOH} \rightarrow \text{CH}_3\text{O} + \text{OH}$; switchover pressures for $k(T,P)/k_\infty(T) = 1/2$ predicted by Eq.(12) vs. CHEMDIS $k(T,P)/k_\infty(T)$ contours.

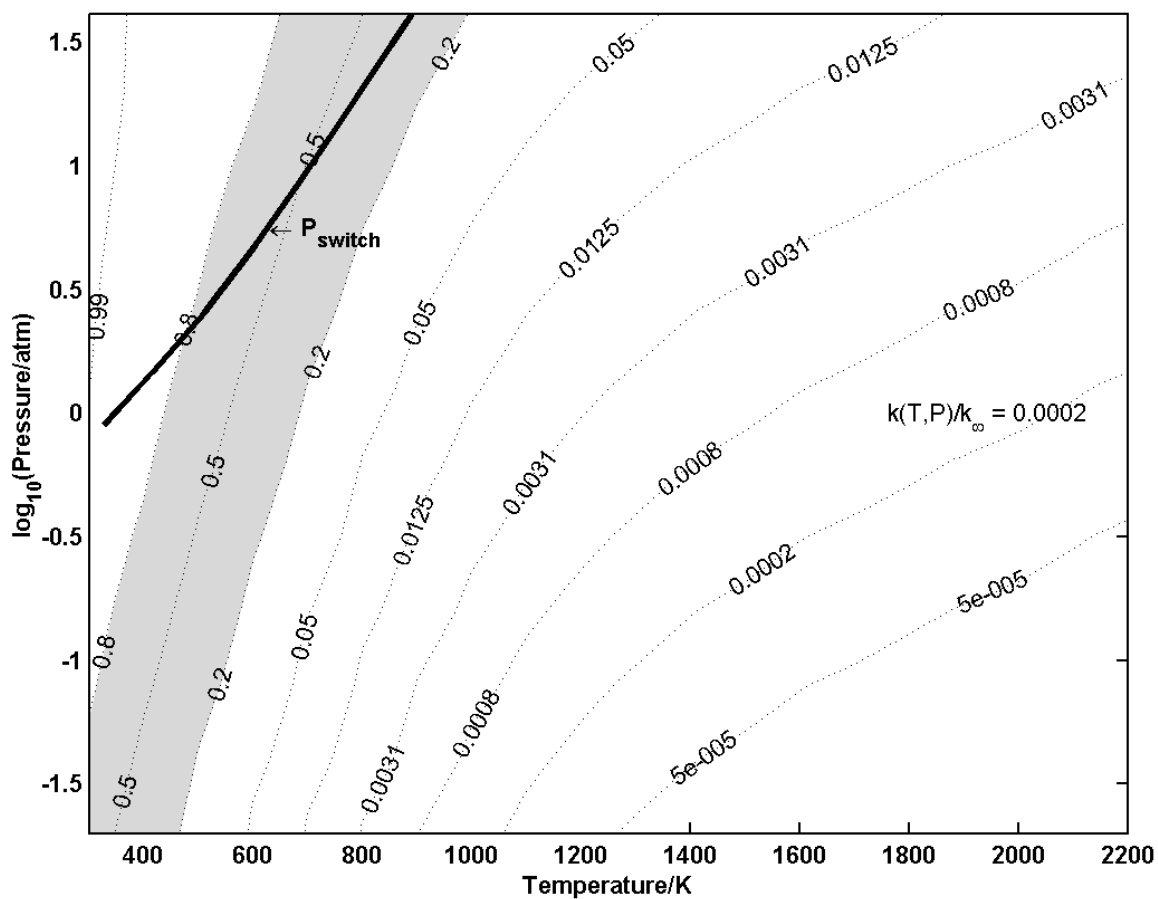


Figure 10. $\text{C}_6\text{H}_6\text{OH} \rightarrow \text{C}_6\text{H}_6 + \text{OH}$; switchover pressures for $k(T,P)/k_{\infty}(T) = 1/2$ predicted by Eq.(12) vs. CHEMDIS $k(T,P)/k_{\infty}(T)$ contours.